

Handover — Read This First

XRayVision AI — FYP documentation package Muhammad Ali Raza (2022F-MUL-BSSWE-017) · Hamza Afzal (2022F-MUL-BSSWE-027) Supervisor: Maam Misbah · School of Software Engineering, Minhaj University Lahore

1. What you have

Folder / file	What it is
XRayVision_AI_Thesis.docx	The submission copy. Formatted for binding.
pdf/	All six documents pre-printed to A4 PDF, page-numbered
01-SRS.md ... 05-Thesis.md	The source documents — edit these
*.html	Same documents for reading in a browser, diagrams already rendered
diagrams/	22 diagrams as SVG (vector) and PNG (print) — paste into slides or a poster
screenshots/	19 screenshots from the live app
tools/	Scripts that regenerate everything
README.md	Full index — read it after this page

2. Do these five things

1. **Open** XRayVision_AI_Thesis.docx **in Word**. Click the **Table of Contents** and press **F9** → *Update entire table*. The contents page is intentionally empty until you do this — Word fills in the page numbers.
2. **Sign the Certificate of Approval and the Declaration of Originality**. The signature blocks are blank and waiting.
3. **Compare against your department's thesis template**. Certificate wording, logo placement and font rules differ between departments. Everything here is plain Word formatting, so adjust freely.
4. **Read Chapter 6 before your viva**. It reports that 17 of 93 test cases were executed and that three defects were found. Two are now fixed; one (DEF-03) is not. Know what it says — an examiner who reads it will ask.

5. **Fix DEF-03 if you can.** Root cause is identified in Chapter 6: the deployed backend was loading the generic COCO checkpoint `yolo8n.pt` (which has a `vase` class) instead of `models/fracture_yolo8n.pt`. Correct `YOLO_WEIGHTS_PATH` on the deployed Space.
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3. Do not do these

- **Do not edit the `.docx` and the `.md` both.** The `.md` files are the source; the `.docx`, `.html` and PDFs are generated from them. If you edit the Word file directly, stop rebuilding — or your edits get overwritten. Pick one and stick to it.
 - **Do not panic if the diagrams look blank in VS Code.** VS Code cannot render Mermaid on its own. Open the `.html` file instead, or install the free extension `bierner.markdown-mermaid`.
 - **Do not submit `pdf/05-Thesis-browser-edition.pdf` as the thesis.** It is the browser version — no binding margin, no paginated contents. Export the real PDF from the `.docx` after step 2.1 above.
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4. If you change the documents

```
cd tools
pip install playwright && python -m playwright install chromium
npm install

python build_html.py           # Markdown -> HTML + diagrams/*.svg
python render_diagrams_png.py  # SVG -> print-quality PNG
node build_thesis_docx.js      # Thesis Markdown -> Word
python build_pdf.py            # HTML -> A4 PDFs
```

Then repeat step 2.1 (F9 in Word). Full details in [tools/README.md](#).

5. Honest status

The documents describe the system accurately, and where something is unverified or broken they say so rather than glossing over it:

- **93 test cases are specified; 17 were executed** against the live deployment, with screenshot evidence. The other 76 are written up and ready to run, but they have **not** been run. Do not present them as passing.
- **Three defects were found.** DEF-01 (`.env` tracked in git) and DEF-02 (backend README contradicting the code) are fixed. **DEF-03 is open** — see item 5 above.
- **Diagnostic accuracy was never validated.** The AI models are third-party pretrained artefacts used as supplied. The thesis makes no clinical accuracy claim, and you should not make one either.

If you run the remaining tests or fix DEF-03, update Chapter 6 and §16.1 of the test document, then rebuild.

Educational use only. XRayVision AI is not a certified medical device.